Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	McCulloch, Andrew D Ph.D.		
Project Title:	Integrated Modeling of Cardiac Mechanical and Electrical Function		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	La Jolla	**State:**	CA
Zip Code:	92093-0412	**Congressional District:**	49
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	04/01/2001	**End Date:**	07/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Omens, Jeffrey (UCSD) Michailova, Anushka (UCSD)		
Grant/Contract No.:	NCC 9-58-CA.002.16		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Aim 1: To develop anatomically detailed dynamic finite element models of three-dimensional cardiac electromechanics. Aim 2: To use these models to bridge systems models cellular dynamics and signaling with systems models of hemodynamics and cardiovascular regulation Aim 3: To apply these models to investigate cardiac responses to chronically altered external loading conditions and develop simulations of cardiac adaptation to microgravity. Aim 4: To implement the models using modular object-oriented software engineering techniques that allow them to be readily integrated and used interactively Aim 5: To collaborate with other projects in the integrative human function core including: Drs Kushmerick and Coolahan, Bers, Rice, Cabrera, and Chase.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many of the most critical health risks of space flight and long-term exposure to microgravity involve cardiac mechanical and electrical function, especially impaired responses to orthostatic and exercise stress, ventricular atrophy and remodeling, and cardiac arrhythmias. One objective of this proposal is to develop computational models of human cardiac electrical and mechanical responses to microgravity that integrate (i) across scales of organization from cell to organ system; (ii) with other functional subsystems of the heart; and (iii) with other physiological systems. Since the initial insult, microgravity, is fundamentally mechanical and because models must be predictive to be useful for countermeasure development and assessment, the core of the models will be anatomically detailed three-dimensional models based on physical first principles. These continuum models will provide a framework for embedding systems models of cellular dynamics and signal transduction and for coupling the heart with systems models of whole-body metabolism, hemodynamics and physiological control. The project will result in software for anatomically and physically based computational models that bridge systems models of cellular physiology to systems models of whole body physiology. Since the models are based on validated finite element methods for biomechanics and electro-physiology, the software may also have applications to other vital aspects of integrated human function in space such as musculoskeletal performance.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)